



CIVITAS indicators

Congestion – Version 4 (TRA_FC_CG4)

DOMAIN

 Transport	 Environment	 Energy	 Society	 Economy
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TOPIC

Functionality of the transport system

IMPACT

Road congestion

Reducing congestion in the urban area

TRA_FC

Category

Key indicator	Supplementary indicator	State indicator
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CONTEXT AND RELEVANCE

Urban congestion refers to the overcrowding of vehicles on road networks, leading to slower speeds, longer trip times, and increased vehicular queuing. Congestion leads to economic losses due to wasted time and fuel, in addition to contributing to air pollution and climate change. Congestion is influenced by various factors, including population size, density, economic activities, urban planning and design, and the availability and adoption of public transportation.

This indicator provides a measure of the level of congestion in the experiment area. **It is a relevant indicator when the policy action is aimed at reducing the traffic jam and the consequent increase of travel time. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

This indicator is based on the speed of vehicles on a sample of road sections compared to the theoretical free flow speed.

The indicator is based specifically on car speed because it is the most representative and, focusing on one vehicle type only avoids complexities due to the mix of different vehicle types on roads.

The indicator is a ratio; therefore, it is **dimensionless**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously** on the required inputs and then coded in the supporting tool.

There are three alternative methods of calculation available for this indicator. The three methods distinguish for the approach used to measure speed. Approaches are of different complexity and significance.

METHOD 1	METHOD 2	METHOD 3
Speeds extracted from a transport simulation model	Speed measured in peak time of one day	Speed measured in the whole day in different days
It is based on a theoretical calculation rather than on observation.	It is based on the real conditions but not they are not directly observed.	It is based on real conditions directly observed.
Complexity 	Complexity 	Complexity 
Significance 	Significance 	Significance 

Method 1

Speeds extracted from a transport simulation model

Significance: **0.25**



INPUTS

The input needed to compute this indicator is:

- a) A set of modelled speeds on a sample of road sections in the experiment area
- b) The theoretical free-flow speed on the sample roads.

The experiment would be reflected in the modification of the modelled speed **in same road sections**. The theoretical free-flow speed of the roads could also be changed. The modifications would be the result of the simulation of one or more interventions directly or indirectly affecting the speed of road vehicles or the free flow speed.

METHOD OF CALCULATION

The indicator should be computed according to the following steps:

- **Definition of the set of road sections** where measuring speed. The selection of sections should be made according to the following rules:
 - Sections should include the major roads in the experiment area.
 - If the experiment includes interventions on some specific roads, counts should be made in at least some of these roads as well as in at least some roads that could be used as alternative by motorists.
 - If part of the experiment consists of including some roads in a pedestrian area where motorised vehicles are forbidden, these roads should NOT be included in the sample.
 - If part of the experiment consists of restricting access to some roads (e.g. low emissions zone) these roads can be included in the sample, but a sufficient number of roads outside the restricted area should also be included.
- **Extraction of modelled speed on each section and free-flow speeds from a transport model.**
- **Classification of sections** between congested and uncongested.
- **Quantification of the indicator.** The indicator is the share of sections where the modelled speed is above the 120% of free-flow speed.

EQUATIONS

The equations that should be applied to quantify the indicator are the followings:

Classification of sections:

$$\begin{aligned} CongLev_s &= 1 & \text{if } Speed_s \geq FFSpeed_s * 1.2 \\ CongLev_s &= 0 & \text{otherwise} \end{aligned}$$

Where:

$Speed_s$ = modelled speed (in km/h) on road section s

$FFSpeed_s$ = Free-flow speed (in km/h) on road section s

Computation of the index:

$$CongInd = \frac{\sum_s CongLev_s}{S}$$

Where:

S = Total number of road sections

Method 2

Speed measured in peak time of one day

Significance: **0.50**



INPUTS

The input needed to compute this indicator is:

- A set of measured speeds of vehicles on a sample of road sections in the experiment area.
- The theoretical free-flow speed on the sample roads.

The experiment would be reflected in the modification of the measured speeds **in same road sections**. The theoretical free-flow speed of the roads could also be changed. The modification would be the result of one or more interventions directly or indirectly affecting the speed of road vehicles.

METHOD OF CALCULATION

The indicator should be computed according to the following steps:

- Definition of the set of road sections** where measuring speed. The selection of sections should be made according to the following rules:
 - Sections should include the major roads in the experiment area.
 - If the experiment includes interventions on some specific roads, counts should be made in at least some of these roads as well as in at least some roads that could be used as alternative by motorists.
 - If part of the experiment consists of including some roads in a pedestrian area where motorised vehicles are forbidden, these roads should NOT be included in the sample.
 - If part of the experiment consists of restricting access to some roads (e.g. low emissions zone) these roads can be included in the sample, but a sufficient number of roads outside the restricted area should also be included.

If in the experiment area there is already a speed monitoring system in place, the data from the system can be used, but the requirements above should be respected anyway.

- **Measurement of vehicles speed on each section.** The measurement should be made in 2 peak time hours of a working day.
- **Calculation of average speed in each section.**
- **Classification of sections** between congested and uncongested.
- **Quantification of the indicator.** The indicator is the share of sections where the modelled speed is above the 120% of free-flow speed.

EQUATIONS

The equations that should be applied to quantify the indicator are the followings:

Calculation of average speed in each section:

$$AvSpeed_s = \frac{\sum_v vSpeed_s}{V_s}$$

Where:

$vSpeed_s$ = observed speed (in km/h) of vehicle v on road section s

V_s = Number of vehicles for which speed is measured on road section s

Classification of sections:

$$CongLev_s = 1 \quad \text{if } AvSpeed_s \geq FFSpeed_s * 1.2$$

$$CongLev_s = 0 \quad \text{otherwise}$$

Where:

$FFSpeed_s$ = Free-flow speed (in km/h) on road section s

Computation of the index:

$$CongInd = \frac{\sum_s CongLev_s}{S}$$

Where:

S = Total number of road sections

Method 3

Speed measured in the whole day in different days

Significance: 1.00



INPUTS

See Method 2.

METHOD OF CALCULATION

See Method 2.

The difference with respect to Method 2 is in the second step, as there are additional requirements regarding the period of the measurements, which should be repeated for at least 3 working days.

EQUATIONS

The equations that should be applied to quantify the indicator are the followings:

Calculation of average speed in each section:

$${}^dAvSpeed_s = \frac{\sum_v {}^dSpeed_{vs}}{dV}$$

Where:

${}^dSpeed_{vs}$ = observed speed (in km/h) of vehicle v on road section s in day d

dV_s = Number of vehicles for which speed is measured on road section s in day d

Classification of sections:

$${}^dCongLev_s = 1 \quad \text{if } {}^dAvSpeed_s \geq FFSpeed_s * 1.2$$

$${}^dCongLev_s = 0 \quad \text{otherwise}$$

Where:

$FFSpeed_s$ = Free-flow speed (in km/h) on road section s

Computation of the index:

$$CongInd = \frac{\sum_d \sum_s {}^dCongLev_s}{S * D}$$

Where:

S = Total number of road sections

D = Number of days monitored

ALTERNATIVE INDICATORS

Alternative indicators for measuring the same impact are **TRA_FC_CG1**, **TRA_FC_CG2**, **TRA_FC_CG3**.

The indicator described in this factsheet assesses congestion severity by measuring the proportion of road sections where observed speeds fall below the free-flow speed by a specified percentage. This metric helps filter out roads experiencing only mild congestion, focusing instead on those with more significant traffic jams. By doing so, it provides insight into the share of road sections affected by congestion above an acceptable threshold.

Meanwhile, **TRA_FC_CG1** takes as input a set of measurements of travel time by car for a sample of origin/destination pairs in the experiment area. This indicator is to be preferred when the goal is

to capture the actual experience of travellers regarding travel times in the experiment area. The input data is obtained from either simulation models, online route search engines or driving tests.

TRA_FC_CG2 assesses traffic congestion by considering traffic counts on a sample of road sections in the experiment area. Its required inputs tend to be simple to collect using sensors or through manual collection. The interpretation of this indicator requires accounting for road capacity since high volumes of traffic on a corridor do not imply congestion if capacity is sufficient. In addition, it does not assess congestion severity, as it does not consider travel time or speed.

TRA_FC_CG3 evaluates congestion intensity by analysing speed data from a sample of road sections, with lower speeds indicating higher congestion levels. Its input data is sourced from either simulation models or road measurements. Similar to **TRA_FC_CG4**, it relies on input data sourced from either simulation models or road measurements.

Each of the alternative indicators has multiple calculation methods, with varying significance and complexity. Methods that use measures collected from traffic simulation models or online travel planners are simpler but less significant. In contrast, methods using field measurements, especially if obtained over multiple days, are more significant and more complex to apply, given the larger efforts needed for data collection.

Data availability can affect the choice among alternative indicators. This indicator (**TRA_FC_CG4**) and **TRA_FC_CG3** may be preferable for cities where speeds are already monitored on a regular basis. In addition, it should be noted that devices used to measure speeds may also count vehicles; therefore, the input data to compute **TRA_FC_CG2** could be easily obtained while taking the measurements needed for this indicator.

It should also be noted that congestion is one of the three components of **TRA_FC_AC1**. This indicator combines public transport connectivity, bike connectivity and road congestion to assess overall accessibility. The online tool automatically computes **TRA_FC_AC1** if the three sub-indicators have been calculated.

Lastly, **TRA_FC_PCG** measures perceived road congestion within the experiment area. This indicator reflects social norms, routines, expectations, and the psychological impact of traffic conditions, rather than providing an objective assessment of the state of the road network.